

NOT ALL LANGUAGE MODEL FEATURES ARE ONE-Dimensionally LINEAR

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ABSTRACT

Recent work has proposed that language models perform computation by manipulating one-dimensional representations of concepts (“features”) in activation space. In contrast, we explore whether some language model representations may be inherently multi-dimensional. We begin by developing a rigorous definition of irreducible multi-dimensional features based on whether they can be decomposed into either independent or non-co-occurring lower-dimensional features. Motivated by these definitions, we design a scalable method that uses sparse autoencoders to automatically find multi-dimensional features in GPT-2 and Mistral 7B. These auto-discovered features include strikingly interpretable examples, e.g. *circular* features representing days of the week and months of the year. We identify tasks where these exact circles are used to solve computational problems involving modular arithmetic in days of the week and months of the year. Next, we provide evidence that these circular features are indeed the fundamental unit of computation in these tasks with intervention experiments on Mistral 7B and Llama 3 8B, and we examine the continuity of the days of the week feature in Mistral 7B. Overall, our work argues that understanding multi-dimensional features is necessary to mechanistically decompose some model behaviors.

1 INTRODUCTION

Language models trained for next-token prediction on large text corpora have demonstrated remarkable capabilities, including coding, reasoning, and in-context learning (Bubeck et al., 2023; Achiam et al., 2023; Anthropic, 2024; Team et al., 2023). However, the specific algorithms models learn to achieve these capabilities remain largely a mystery to researchers; we do not understand how language models write poetry. Mechanistic interpretability is a field that seeks to address this gap by reverse-engineering trained models from the ground up into variables (features) and the programs (circuits) that process these variables (Olah et al., 2020).

One mechanistic interpretability research direction has focused on understanding toy models in detail. This work has found multi-dimensional representations of inputs such as lattices (Michaud et al., 2024) and circles (Liu et al., 2022; Nanda et al., 2023a), and has successfully reverse-engineered the algorithms that models use to manipulate these representations. A separate direction has identified one-dimensional representations of high level concepts and quantities in large language models (Gurnee & Tegmark, 2023; Marks & Tegmark, 2023; Heinzerling & Inui, 2024; Bricken et al., 2023). These findings have led to the *linear representation hypothesis* (LRH): a hypothesis which has historically claimed both that 1. all representations in pretrained large language models lie along one-dimensional lines, and 2. model states are a simple sparse sum of these representations (Park et al., 2023; Bricken et al., 2023). In this work, we specifically call into question the first part of the LRH.¹

¹An earlier version of this manuscript sparked discussion in the mechanistic interpretability community on the distinction between non-linear features and multi-dimensional features, and in fact this discussion directly led

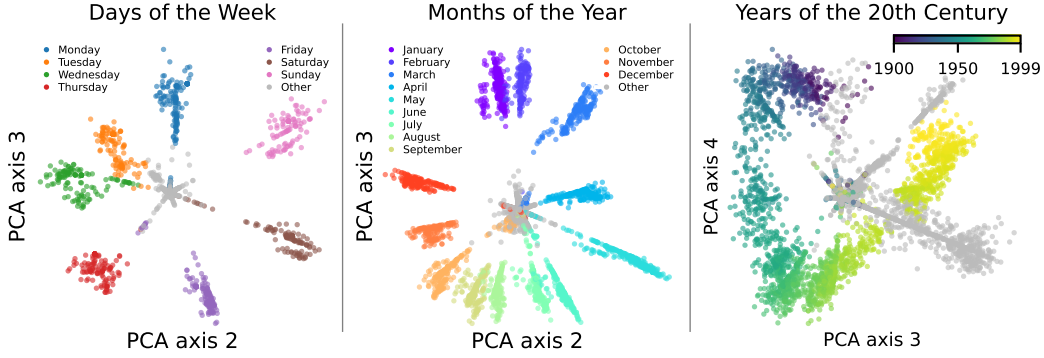


Figure 1: Circular representations of days of the week, months of the year, and years of the 20th century in layer 7 of GPT-2-small colored by the token they fire on. These representations were discovered via clustering SAE dictionary elements, described in Section 4. Points are colored according to the token which created the representation. See Fig. 14 for other axes and Fig. 15 for similar plots for Mistral 7B.

For the most part, these two directions have been disconnected: Yedidia (2023a) and Gould et al. (2023) find intriguing hints of circular language model representations, and Bricken et al. (2023) speculate about the existence of feature manifolds, but these brief results only serve to further emphasize the lack of a unifying and satisfying perspective on the nature of language model features. In this work, we seek to bridge this gap by formalizing, investigating, and systematically searching for multi-dimensional language model features.

1.1 CONTRIBUTIONS

1. In Section 3, we generalize the one-dimensional definition of a language model feature to multi-dimensional features and provide an updated multi-dimensional superposition hypothesis to account for these new features.
2. In Section 4, we build on the definitions proposed in Section 3 to develop a theoretically grounded and empirically practical test that uses sparse autoencoders to find irreducible features. Using this test, we identify multi-dimensional representations automatically in GPT-2 and Mistral 7B, including circular representations for the day of the week and month of the year.
3. In Section 5, we show that Mistral 7B and Llama 3 8B use these circular representations when performing modular addition in days of the week and in months of the year. To the best of our knowledge, we are the first to find causal circular representations of concepts in a language model. We additionally find that the model’s circular representations respect a continuous notion of time.

2 RELATED WORK

Linear Representations: Early word embedding methods such as GloVe and Word2vec, although only trained using co-occurrence data, contained directions in their vector spaces corresponding to semantic concepts (Mikolov et al., 2013b; Pennington et al., 2014; Mikolov et al., 2013a). Recent research has found similar evidence of one-dimensional linear representations in sequence models trained only on next token prediction, including Othello board positions (Nanda et al., 2023b; Li et al., 2022), the truth value of assertions (Marks & Tegmark, 2023), and numeric quantities such as longitude, latitude, birth year, and death year (Gurnee & Tegmark, 2023; Heinzerling & Inui, 2024). These results have inspired the linear representation hypothesis (Park et al., 2023; Elhage et al., 2022)

to a consensus around the two part LRH we describe above (see (Olah, 2024; Csordás et al., 2024; Mendel, 2024; Kantamneni & Tegmark, 2025)). To clarify, we agree with these discussions, and believe the multi-dimensional features we find are "linear" in the sense that they are contained in a low-dimensional linear subspace, but "non-linear" in the sense that this low-dimensional subspace is not one-dimensional (and this is the sense we mean in the title).

defined above. Jiang et al. (2024) provide theoretical evidence for this hypothesis, assuming a latent (binary) variable-based model of language. Empirically, Bricken et al. (2023) and Cunningham et al. (2023) successfully use sparse autoencoders to break down a model’s feature space into an over-complete basis of linear features. These works assume that the number of linear features stored in superposition exceeds the model dimensionality (Elhage et al., 2022).

Multi-Dimensional Representations: There has been comparatively little research on multi-dimensional features in language models. Shai et al. (2024) predict and verify that a transformer trained on a hidden Markov model uses a fractal structure to represent the probability of each next token, a clear example of a necessary multi-dimensional feature, but the analysis is restricted to a toy setting. Yedidia (2023a;b) finds that GPT-2 learned position vectors form a helix, which implies a circle when “viewed” from below. Thus, we are not the first to find a circular feature in a language model. However, our work finds circular features that represent latent concepts from text, while the GPT-2 learned position vectors are specific to tokenization, separate from the rest of the model parameters, and causally implicated only due to positional attention masking. Another suggestive result, due to Hanna et al. (2024), is the presence of a U-shape in the representation of numbers between 0 and 100; however, Hanna et al. (2024) find that this representation is not causal, and they only show it exists within a specific prompt distribution. Recent work on dictionary learning (Bricken et al., 2023) has speculated about multi-dimensional *feature manifolds*; our work is similar to this direction and develops the idea of feature manifolds theoretically and empirically. Finally, in a separate direction, Black et al. (2022) argue for interpreting neural networks through the polytopes they split the input space into, and identifies regions of low polytope density as “valid” regions for a potential linear representation.

Circuits: Circuits research seeks to identify and understand *circuits*, subsets of a model (usually represented as a directed acyclic graph) that explain specific behaviors (Olah et al., 2020). The base units that form a circuit can be layers, neurons (Olah et al., 2020), or sparse autoencoder features (Marks et al., 2024). In the first circuits-style work, Olah et al. (2020) found line features that were combined into curve detection features in the InceptionV1 image model. More recent work has examined language models, for example the indirect object identification circuit in GPT-2 (Wang et al., 2022). Given the difficulty of designing bespoke experiments, there has been increased research in automated circuit discovery methods (Marks et al., 2024; Conmy et al., 2023; Syed et al., 2023).

Interpretability for Arithmetic Problems: Liu et al. (2022) study models trained on modular arithmetic problems $a + b = c \pmod{m}$ and find that models that generalize well have circular representations for a and b . Further work by Nanda et al. (2023a) and Zhong et al. (2024) shows that models use these circular representations to compute c via a “clock” algorithm and a separate “pizza” algorithm. These papers are limited to the case of a small model trained only on modular arithmetic. Another direction has studied how large language models perform basic arithmetic, including a circuits level description of the greater-than operation in GPT-2 (Hanna et al., 2024) and addition in GPT-J (Stolfo et al., 2023). These works find that to perform a computation, models copy pertinent information to the token before the computed result and perform the computation in the subsequent MLP layers. Finally, recent work by Gould et al. (2023) investigates language models’ ability to increment numbers and finds linear features that fire on tokens equivalent modulo 10.

3 DEFINITIONS

This section focuses on hypotheses for how hidden states of language models can be decomposed into sums of functions of the input (features). We focus on L layer transformer models M that take in token input $\mathbf{t} = (t_1, \dots, t_n)$ from input token distribution $\mathcal{T}$, have hidden states $\mathbf{x}_{1,l}, \dots, \mathbf{x}_{n,l}$ for layers l , and output logit vectors $\mathbf{y}_1, \dots, \mathbf{y}_n$. Given a set of inputs T , we let $X_{i,l}$ be the set of all corresponding $\mathbf{x}_{i,l}$. We write matrices in capital bold, vectors and vector valued functions in lowercase bold, and sets in capital non-bold.

3.1 MULTI-DIMENSIONAL FEATURES

Definition 1 (Feature). We define a d_f -dimensional feature as a function $\mathbf{f}$ that maps a subset of the input space into $\mathbb{R}^{d_f}$. We say that a feature is *active* on the aforementioned subset.

The input token distribution $\mathcal{T}$ induces a d_f -dimensional probability distribution over feature vectors $\mathbf{f}(t)$. As an example, let $n = 1$ (so inputs are single tokens) and consider a feature $\mathbf{f}$ that maps integer tokens to their integer values in $\mathbb{R}^1$. Then $\mathbf{f}$ is a 1-dimensional feature that is active on integer tokens, and $\mathbf{f}(t)$ is the marginal integer occurrence distribution from the token distribution.

How can we differentiate "true" multi-dimensional features from sums of lower dimensional features? We make this distinction by examining the *reducibility* of a potential multi-dimensional feature. That is, $\mathbf{f}$ is a "true" multi-dimensional feature if it cannot be written as the sum of two statistically independent features **and** it cannot be written as the sum of two non-co-occurring features. Formally, we have the following definition:

Definition 2. A feature $\mathbf{f}$ is *reducible* into features $\mathbf{a}$ and $\mathbf{b}$ if there exists an affine transformation

$$\mathbf{f} \mapsto \mathbf{R}\mathbf{f} + \mathbf{c} \equiv \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix} \quad (1)$$

for some orthonormal $d_f \times d_f$ matrix $\mathbf{R}$ and additive constant $\mathbf{c}$, such that the transformed feature probability distribution $p(\mathbf{a}, \mathbf{b})$ satisfies at least one of these conditions:

1. p is *separable*, i.e., factorizable as a product of its marginal distributions:
 $p(\mathbf{a}, \mathbf{b}) = p(\mathbf{a})p(\mathbf{b})$.
2. p is a *mixture*, i.e., a sum of disjoint distributions, one of which is lower dimensional:
 $p(\mathbf{a}, \mathbf{b}) = wp(\mathbf{a})\delta(\mathbf{b}) + (1 - w)p(\mathbf{a}, \mathbf{b})$

Here, p is a probability density function that is conditional on the subset of $\mathcal{T}$ that $\mathbf{f}$ is active on, δ is the Dirac delta function, and $0 < w < 1$. By two probability distributions being disjoint, we mean that they have disjoint support (there is no set where both have positive probability measure, or equivalently the two features $\mathbf{a}$ and $\mathbf{b}$ cannot be active at the same time). In Eq. (1), $\mathbf{a}$ is the first k components of the vector $\mathbf{R}\mathbf{f} + \mathbf{c}$ and $\mathbf{b}$ is the remaining $d_f - k$ components. When p is separable or a mixture, we also say that $\mathbf{f}$ is separable or a mixture. We term a feature *irreducible* if it is not reducible, i.e., if no rotation and translation makes it separable or a mixture.

An example of a feature that is a mixture is a one hot encoding along a simplex; an example of a feature that is separable is a normal distribution². In natural language, a mixture might be a one hot encoding of "breed of dog", while a separable distribution might be the "latitude" and "longitude" of location tokens.

In practice, the mixture and separability definitions may not be precisely satisfied. Thus, we soften our definitions to permit degrees of reducibility:

Definition 3 (Separability Index and ϵ -Mixture Index). Consider a feature $\mathbf{f}$. The **separability index** $S(\mathbf{f})$ measures the minimal mutual information between all possible $\mathbf{a}$ and $\mathbf{b}$ defined in Eq. (1):

$$S(\mathbf{f}) \equiv \min I(\mathbf{a}; \mathbf{b}) \quad (2)$$

where I denotes the mutual information. Smaller values of $S(\mathbf{f})$ mean that $\mathbf{f}$ is more separable.

The **ϵ -mixture index** $M_\epsilon(\mathbf{f})$ tests how often $\mathbf{f}$ can be projected near zero while it is active:

$$M_\epsilon(\mathbf{f}) = \max_{\mathbf{v} \in \mathbb{R}^{d_f}, c \in \mathbb{R}} \mathbb{P}_{\mathbf{t} \in \mathcal{T}} \left(|\mathbf{v} \cdot \mathbf{f}(\mathbf{t}) + c| < \epsilon \sqrt{\mathbb{E}[(\mathbf{v} \cdot \mathbf{f}(\mathbf{t}) + c)^2]} \right) \quad (3)$$

Larger values of $M_\epsilon(\mathbf{f})$ mean that $\mathbf{f}$ is more of a mixture.

In Appendix B, we expand on the intuition behind why the separability and ϵ -mixture indices as defined here correspond to weakened versions of Definition 2.

We develop optimization procedures to empirically solve for the separability and ϵ -mixture indices of two dimensional feature distributions. At a high level, the separability procedure iterates over a sweep of rotations and estimates the mutual information between the axes for each angle, while the ϵ -mixture index procedure performs gradient descent to find the ϵ band that contains the largest possible fraction of the feature distribution. For more details on the implementation of the tests, see Appendix B.2. In Section 4, we apply these empirical tests to real language model feature distributions to find irreducible multi-dimensional features; we show the detailed test results on the "days of the week" cluster in Fig. 2

²since any multidimensional Gaussian can be rotated to have a diagonal covariance matrix

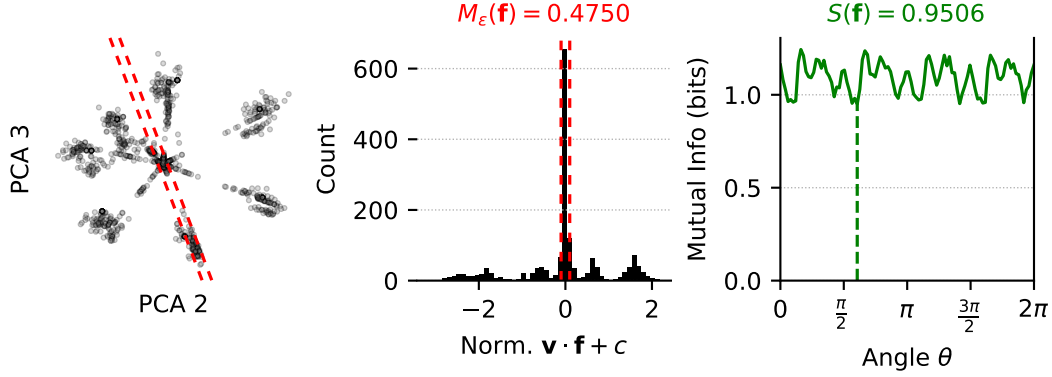


Figure 2: Empirical ϵ -mixture index and separability index for the “days of the week” cluster along PCA components 2 and 3. **Left:** The ϵ band parameterized by $\mathbf{v}$ and c that the optimization procedure found contained the highest fraction of points. **Mid:** Dot products of points in the feature distribution with the ϵ band; $M_\epsilon(\mathbf{f})$ is the percent of dot products within $\epsilon = 0.1$ of 0. **Right:** Estimated mutual information for different rotations of the space; $S(\mathbf{f})$ is the minimum over all rotations. This point cloud has a lower ϵ -mixture index and higher separability index than PCA projections within typical clusters (see Fig. 3), indicating that it is more likely to be an irreducible multi-dimensional feature.

3.2 SUPERPOSITION

In this section, we propose an updated *superposition hypothesis* (Elhage et al., 2022) that takes into account multi-dimensional features. First, we restate the original superposition hypothesis:

Definition 4 (δ -orthogonal matrices). Two matrices $\mathbf{A}_1 \in \mathbb{R}^{d \times d_1}$ and $\mathbf{A}_2 \in \mathbb{R}^{d \times d_2}$ are δ -orthogonal if $|\mathbf{x}_1 \cdot \mathbf{x}_2| \leq \delta$ for all unit vectors $\mathbf{x}_1 \in \text{colspace}(\mathbf{A}_1)$ and $\mathbf{x}_2 \in \text{colspace}(\mathbf{A}_2)$.

Hypothesis 1 (One-Dimensional Superposition Hypothesis, paraphrased from (Elhage et al., 2022)). Hidden states $\mathbf{x}_{i,l}$ are the sum of many ($\gg d$) sparse one-dimensional features f_i and pairwise δ -orthogonal vectors $\mathbf{v}_i$ such that $\mathbf{x}_{i,l}(t) = \sum_i \mathbf{v}_i f_i(t)$. We set $f_i(t)$ to zero when t is outside the domain of f_i .

In contrast, our new superposition hypothesis posits independence between irreducible multi-dimensional features instead of unknown levels of independence between one-dimensional features:

Hypothesis 2 (Multi-Dimensional Superposition Hypothesis, changes underlined). Hidden states $\mathbf{x}_{i,l}$ are the sum of many ($\gg d$) sparse low-dimensional irreducible features $\mathbf{f}_i$ and pairwise δ -orthogonal matrices $\mathbf{V}_i \in \mathbb{R}^{d \times d_{\epsilon_i}}$ such that $\mathbf{x}_{i,l}(t) = \sum_i \mathbf{V}_i \mathbf{f}_i(t)$. We set $\mathbf{f}_i(t)$ to zero when t is outside the domain of $\mathbf{f}_i$.

Note that since multi-dimensional features can be written as the sums of projections of lower-dimensional features, our new superposition hypothesis is a stricter version of Hypothesis 1. In the next section, we will explore empirical evidence for our hypothesis, while in Appendix A, we prove upper and lower bounds on the number of δ -almost orthogonal matrices $\mathbf{V}_i$ that can be packed into d dimensional space.

4 SPARSE AUTOENCODERS FIND MULTI-DIMENSIONAL FEATURES

In this section, we describe a method to identify multi-dimensional features in language model hidden states using sparse autoencoders (SAEs). Sparse autoencoders (SAEs) deconstruct model hidden states into sparse vector sums from an over-complete basis (Bricken et al., 2023; Cunningham et al., 2023). For hidden states $X_{i,l}$, a one-layer SAE of size m with sparsity penalty λ minimizes the following dictionary learning loss (Bricken et al., 2023; Cunningham et al., 2023):

$$\text{DL}(X_{i,l}) = \arg \min_{\mathbf{E} \in \mathbb{R}^{m \times d}, \mathbf{D} \in \mathbb{R}^{d \times m}} \sum_{\mathbf{x}_{i,l} \in X_{i,l}} \left[\|\mathbf{x}_{i,l} - \mathbf{D} \cdot \text{ReLU}(\mathbf{E} \cdot \mathbf{x}_{i,l})\|_2^2 + \lambda \|\text{ReLU}(\mathbf{E} \cdot \mathbf{x}_{i,l})\|_0 \right] \quad (4)$$